

CO Forecasting Architecture Decision Matrix

Model	Temporal Fit 30%	On-Device Latency 25%	Model Size 15%	Training Stability 10%	Impl. Maturity 10%	Interp. 10%	Weighted Score
GRU	4.50 (gated memory)	4.00 (~5 ms)	4.00 (fewer gates)	4.00 (stable)	4.50 (standard RNN)	2.50 (hidden state)	4.050
1D CNN	3.50 (local windows)	4.50 (~2 ms)	4.00 (small)	4.00 (stable)	4.00 (standard conv)	3.00 (filters)	3.875
Dense Baseline	2.00 (no memory)	5.00 (<1 ms)	5.00 (smallest)	4.50 (stable)	4.00 (standard layers)	4.00 (fully connected)	3.850
Temporal CNN (TCN)	4.00 (dilated history)	4.00 (~3 ms)	3.50 (medium)	4.00 (stable)	3.50 (custom blocks)	3.00 (filters)	3.775
LSTM	4.50 (gated memory)	3.00 (~8 ms)	3.00 (largest)	3.50 (less stable)	4.50 (standard RNN)	2.50 (hidden state)	3.600

Scores are normalized on a 1-5 scale; bracketed values show the measured value or scoring basis.